

Robot & grasp geometry

Object + goal + assigned rails



2×7 -DoF arms

Dexterous hands
Lift + torso

Holonomic base



Grasp frame $\bar{T}_{ih}$

Clearance + IK

Shared planning



RRT-Connect

MPC · 10 Hz

Spatial schedule: $|\phi| \leq 90^\circ$

Acquisition supervisor

Shared local actor · 50 Hz

Own sensor history + references

Joints / fingers · object / hand errors
Local load / contact · sensor age

A · MABEL



B · MILO



Same MLP weights; separate histories
 $8 \times 352 \rightarrow 128 \rightarrow 128 \rightarrow 23$

Actuation + contact

23 actions / robot

Base (3) · lift (1) · torso (1)
Arms / wrists (16)
Gripper closure (2)

Motor limits + servos

MuJoCo · 500 Hz

Finger-rail friction contacts
Measured motion / forces

Central training only · PPO

Rollouts + rewards
Pose / grasp / stability

Privileged critic
Shared actor update

Critic uses simulator state; actors use local history

Full-mission baseline · separate actor and grasp model



SE(2) RRT $\rightarrow$ MPC (10 Hz) $\rightarrow$
21-action local actor

Imitation / DAgger · phase and height supervisor

Contact-gated welds · MuJoCo 200 Hz

Measured object / hand poses, encoders and local contact forces

Training experience

pose / twist reference

baseline actions